Jasper Sands

Palo Alto, CA | +1 (650) 924-8429 | jaspersands02@gmail.com | jaspersands.com | github.com/Jaspersands | linkedin.com/in/jaspersands

Quantum error correction, fault-tolerant quantum algorithms, and machine learning for quantum-device characterization. MS in Computer Science at Columbia. Thesis at Diraq and UNSW Sydney on automating silicon spin-qubit measurement.

EDUCATION

Columbia University, MS in Computer Science, Quantum Computing Track · GPA 3.95/4.0 Sep 2025 – May 2027
Quantum coursework: Error Correction · Engineering · Simulation & Computing Lab · Optimization & ML.
TA: Introduction to Quantum Computing.

Washington University in St. Louis, BS in Computer Science and Economics · GPA 3.81/4.0 Aug 2021 – May 2025
Coursework: Machine Learning · Analysis of Algorithms · Computer Security. **TA:** Malware Analysis; Parallel Programming.

SELECTED PROJECTS

stabilizer-qec: surface-code simulator and decoder · *Rust, Python, WebAssembly* qcompiler.jaspersands.com
- Circuit-level simulator and decoders for surface, XZZX and IBM gross codes; matches Stim and PyMatching shot for shot.
- Rebuilt Google's Willow and Sycamore experiments (27.5M shots) from raw measurements bit for bit and fitted Λ as Google does.

ft-algorithms: algorithms on a simulated fault-tolerant quantum computer · *Rust, Python* ft.jaspersands.com
- Runs Shor ($N \leq 143$) and H_2/LiH phase estimation on a lattice-surgery machine fitted to 316 circuit-level QEC experiments.
- Reproduces Gidney's 2025 RSA-2048 distance ($d = 25$) under his noise projection; measured noise needs $d = 27$–29.

Q-Search: proof-gated quantum algorithm research and verification engine qsearch.jaspersands.com
- Automated search for structural quantum advantage on non-abelian HSP, dihedral coset, and linear code equivalence problems.
- Formal proof gates test each mechanism against 1,200+ classical baselines across 720+ theorem modules, indexing negative results.

WhatTheDuck: quantum amplitude estimation for financial Value at Risk · *CUDA-Q, C++* MIT iQuHACK 2026
- Wrote the GPU-accelerated state prep, threshold oracle, and bisection search driving Iterative Quantum Amplitude Estimation.
- Measured query scaling matched the $O(1/\varepsilon)$ bound against an $O(1/\varepsilon^2)$ quasi-Monte Carlo classical baseline.

RESEARCH

Diraq / UNSW Sydney · *MS Thesis Research* Sep 2026 – May 2027
- Automating silicon spin-qubit characterization under Dr. Nard Dumoulin Stuyck, replacing manual tuning and interpretation.
- Acquiring charge measurements from cryogenic probing, and classifying device states against what an expert would call by hand.

Cidon Systems Research Lab, Columbia University · *Graduate Researcher* Sep 2025 – Jun 2026
- Built phishing and spam detection with Dr. Asaf Cidon and Barracuda Networks from LLM embeddings and email telemetry.

Complex Resilient Intelligent Systems Lab, Columbia University · *Graduate Researcher* May – Dec 2025
- Reduced hallucination in generative protein modeling under Dr. Venkat Venkatasubramanian, grounded in molecular graph data.

Foster Care Policy Database, Washington University in St. Louis · *Undergraduate Researcher* Sep 2024 – May 2025
- Fine-tuned LLaMA-3 with Hugging Face for QA over 50k foster-care policy documents, with human-in-the-loop feedback.

EXPERIENCE

Smack Technologies · *AI Intern* May – Aug 2026
- Built physics and sensor simulation, integrating the knowledge graph to expand corpus generation.
- Benchmarked open-weight models head to head on speed, intelligence and concurrency, speeding corpus generation 64x.
- Bounded simulator error to 0.55% against references outside its own tests, in a report that regenerates when the physics changes.

Highnote · *Cybersecurity Engineer Intern* May – Jul 2024
- Built the company-wide SIEM for a card-issuing fintech: AWS, GCP and Datadog logs in one Elasticsearch cluster (1 TB/day), 100+ detections.

Mindtrip · *Software Engineer Intern* May – Jul 2023
- Built internal admin and secure-query tooling in JavaScript and Rails via Retool, saving roughly 250 engineering hours a month.

COMPETITIONS & LEADERSHIP

- **Overall Winner** & **NVIDIA Ecosystem Award**, MIT iQuHACK 2026, qualified for the NYU Abu Dhabi Quantum and AI Hackathon 2027 · **Audience Favorite**, Harmoniqs Quantum Design 2026 · **Runner-Up**, Qualcomm Snapdragon Hackathon 2026
- **Graduate Lead**, Columbia Quantum Algorithms Reading Group: weekly sessions through all 33 chapters of Andrew Childs' notes.

SKILLS

Quantum tools: Qiskit and Qiskit Runtime, Cirq, PennyLane, CUDA-Q, Stim, PyMatching, QuTiP
Quantum methods: Hamiltonian simulation, phase and amplitude estimation, QSVT and block encoding, VQE, QAOA · surface codes, stabilizer formalism, MWPM and Union-Find decoding, threshold estimation, noise modeling, lattice surgery
Languages: Python, C++, Rust, C, MATLAB, JavaScript, TypeScript, Go, Java
ML & systems: PyTorch, CUDA, Hugging Face · Linux, Git, Docker, WebAssembly, AWS, GCP, Elasticsearch, LaTeX